A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 26

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Extensive:** low inputs of energy, food and chemicals so low cost and low environmental impact – sustainable – but low yield per area and dependent on natural productivity **up to 2 Intensive:** control of all limiting factors gives very high yield and profit per area – but high inputs (fishmeal from wild fish, energy), disease and lice at high density, pollution from wastes, antibiotic / pesticide use, escapes **up to 2** Judgement depends on species, market and site **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Adults with desirable characteristics (fast growth, shape, colour) stripped of eggs and milt which are mixed for fertilisation **1** Eggs raised in aerated freshwater tanks, dead / diseased eggs removed daily – fry fed fishmeal pellets and moved through tanks as they grow **1** After 12-18 months smolts transferred to seawater cages, fed pellets until harvest at 3-5 kg **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**1 mark each** 1 mark for each of the 3 points below. **1** About 10% of the world's mangroves have been cleared for coastal shrimp lagoons (up to 100 ha), removing nurseries and coastal protection **1** Lagoons need water exchange to remove wastes and aeration to prevent deoxygenation as wastes decompose **1** Eyestalk ablation stimulates breeding – high densities increase disease so antibiotics / pesticides used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-356; AQA spec 3.5.2.3.	
02.1	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Potato bean (N America): high protein content **1** Morama bean (southern Africa): protein content similar to soya, drought resistant **1** Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils **1** Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion **1 American bison:** graze grassland too poor for cattle **1 Common eland:** browse and graze so can be kept with cattle without severe competition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	2

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil **1** So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	1 mark for the point below. **1** They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	
03.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Ground surface:** pitfall trap (container sunk level with the surface, cover, checked after 24 h) - only mobile surface animals – trapped predators eat others – preservative attracts / repels – escapes **2 Leaf litter:** Tullgren funnel (sample on mesh under a light; heat / light drive invertebrates down into a container for a standard time) - only species repelled by heat / light – some die or stop moving – large ones cannot pass the mesh **2 Branches:** beating tray (branch beaten a set number of times over a sheet) - hard to standardise vigour – some species not dislodged / fly away – high branches unreachable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pitfall trap** → a sunken container that catches ground invertebrates • **Tullgren** → a funnel that drives soil invertebrates down into a collecting jar	6
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Moths only fly as adults for part of the year and activity depends on weather / season, so a species may be present but not caught **1** Brightness / wavelength of the lamp affects attractiveness so catches vary between traps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** The area of ground covered by the suction pipe is known so invertebrates per m^2 can be calculated **1** A pooter is only a handling / collection device for individuals already found, not a sampling method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 397-399; AQA spec 3.7.2.1.	
04.1	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Success:** Cactoblastis moth introduced from South America controlled prickly pear cactus in Australia – Encarsia parasitic wasps control greenhouse whitefly **up to 2 Problems:** cane toads introduced to Australia to control sugar cane beetles did not control them but ate many other species and are toxic to predators – rosy wolfsnail introduced to Polynesia to control African land snails ate endemic Partulid snails (48 species extinct) – harlequin ladybirds introduced to Europe eat native ladybirds **up to 2 Lesson:** control agents should be specialist feeders on the pest **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest • **endemic** → found naturally only in that one place	4
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Release an artificial sex scent attracting males (or females) of the pest species **1** Used to monitor when the pest arrives so pesticide is timed correctly, or (in enclosed greenhouses) to trap enough of one sex to prevent breeding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Regular low doses kill only the most sensitive bacteria so resistant strains survive and multiply **1** Agricultural use exceeds medical use, so resistance spreads **1** Resistant zoonotic bacteria (E. coli, Salmonella, Campylobacter) can transfer to humans and are harder to treat – hence reduced use of antibiotics is a sustainability goal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
04.4	1 mark for the point below. **1** Grow nectar-rich flowering plants alongside crops – restrict pesticides harmful to bees / spray when not in flower – introduce bee hives **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 316-320; AQA spec 3.5.1.1.	
05.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Demersal trawl:** seabed species - cod, haddock, plaice, shrimps, scampi – chains / weights damage the seabed (mix aerobic and anaerobic layers, destroy sea fans, corals, seagrass) – high by-catch in mixed fisheries – high fuel use **2 Purse seine:** shoaling pelagic species - tuna, sardines, herring, anchovy – can remove whole shoals – dolphins associated with tuna shoals may be caught **2 Pelagic long-line:** tuna, squid – albatrosses (long-lived, low breeding rate) drown on baited hooks – sharks and turtles caught **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Shrimp trawls use very small mesh so crabs, molluscs, sea urchins, starfish, juvenile fish cannot escape - by-catch can be 99% of the catch **1** Drift nets are non-selective curtains near the surface catching whales, dolphins, turtles, sharks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Lost or discarded nets and traps continue to catch and kill organisms – dead animals act as bait attracting more **1** Biodegradable ropes / panels so lost traps fall apart – radio transmitters on traps so they can be recovered – net recovery schemes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biodegradable** → broken down by decomposers **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 340-343; AQA spec 3.5.2.2.	
06.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 Tracking:** continuous data on movements, territory size, migration routes and habitat use without disturbance, over large areas and long periods – acoustic tags in seawater – geolocators for small birds – more representative than occasional sightings **2 DNA:** identifies individuals, gene pool size / inbreeding and regional populations (tracing ivory, timber, fish) – eDNA detects rare or hidden aquatic species (great crested newt, invasive carp) from shed cells without capture **2 Image recognition:** software identifies individuals from unique markings (tiger stripes, whale shark spots, fin damage) in camera-trap / drone / satellite images, giving population size, lifespan, social structure from large image sets quickly and objectively **2 Acoustic:** bat detectors, hydrophones and sonograms record species that are hard to see (bats, cetaceans, insects), giving presence, abundance and activity continuously, day and night **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **validity** → actually measures what it is meant to measure • **gene pool** → the range of alleles / genes present in a population	8

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Data collected continuously / over large areas / at night rather than at a few times and places a researcher can visit **1** Less disturbance so animals behave naturally – larger sample sizes – objective rather than subjective identification **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-406; AQA spec 3.1.2.3.6.	
07.1	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Hydrological cycle **1** Carbon cycle **1** Nitrogen cycle **1** Atmospheric circulation **1** Thermohaline circulation **1** Photosynthesis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'plants breathe in carbon dioxide and breathe out oxygen' - name photosynthesis and respiration	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Enzymes reduce the activation energy of reactions **1** So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	2
07.3	**Any 1 of 3** 1 mark each for any 1 of the 3 points below, in any order, maximum 1. **1** Haber process (nitrogen fixation) **1** Incineration of wastes (decomposition) **1** Many manufacturing processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **Incineration** → burning waste, usually with energy recovery	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 3 points below. **1 Hard release:** individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) **1 Soft release:** gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds **1 Problems faced:** finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
08.2	**Any 5 of 7** 1 mark each for any 5 of the 7 points below, in any order, maximum 5. **1** Large enough suitable habitat for a viable population **1** Reliable food supply **1** Low predation risk / original threat removed **1** Suitable breeding sites **1** Water **1** Support of the local human population **1** Official / legal protection of habitat and species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	5
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed **1** Identifies causes of failure so future releases can be improved / extra support given **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 4 points below. **1** No major temperature extremes – regular water supply with no pronounced dry season – deep fertile soils built by leaf litter and root binding – strong seasonality **1** Deciduous canopy present spring to autumn shades the floor during the best growing temperatures **1** Bluebells, snowdrops, lesser celandine grow and flower early in spring before the trees leaf **1** Winter food shortage causes animals to hibernate (dormice), store food or migrate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Long time for species to colonise – some woodland plants colonise very slowly so their presence (ancient woodland indicators) shows the wood is old **1** Natural woodland has trees of all ages with clearings where old trees died, giving varied light and dead wood habitats **1** A new plantation has a single-age dense canopy with low light and few niches until trees start to die (over 200 years) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	3
09.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Clearance for farmland, plantations, urban development, roads, mineral extraction **1** Fragmentation isolating remaining woods so populations cannot move between them and are vulnerable **1 Management change:** end of coppicing so woods become uniform and dark – replacement by conifer monocultures **1** Overgrazing by deer preventing regeneration – introduced pathogens (ash dieback) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 10% of land is already cultivated and 25% grazed **1** Most remaining land is too dry, wet, cold, hot or steep, is covered in ice, or has no soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Land becomes too expensive for local people to rent / is used to grow export crops for more affluent societies **1** So people with less money lose access to the land and food they used to produce **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** It uses natural processes to supply nutrients (legumes, recycled organic matter, soil biota), control pests and provide ecological services **1** So that pesticides and artificial fertilisers become unnecessary rather than just being banned **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Unsustainable features:** reliance on chemical pesticides (resistance, non-target impacts, bans), non-renewable rock phosphate, fossil-fuel-intensive nitrate manufacture, over-exploitation of rivers / groundwater, salinisation, loss of soil organic matter, greenhouse gases from fuel, rice and livestock, loss of species providing ecosystem services • **Pest control:** cultural methods (weeding, mulching, crop rotation, barrier crops, culling), biological control, predator habitats, polyculture / companion crops, integrated control, reduced antibiotic use • **Nutrients:** recycling organic matter, crop rotation, cultivation of legumes, conservation of soil biota, reduced artificial fertiliser especially nitrate, low tillage • **Water:** low water-use crops, reservoirs and aquifer recharge, drip irrigation rather than sprays • **Soil and carbon:** low-tillage farming to maintain organic matter, reduced machinery, renewable energy, drought-tolerant rice varieties so fields are drained earlier, high-carbohydrate ground feed for cattle reduces methane • **Biodiversity:** conserve habitats of crop wild relatives, seed banks, retain hedgerows, ditches, ponds, woodland – agri-environment schemes • **Evaluation:** growing population (perhaps 12 billion) needs more food – lower trophic level diets – obstacles - cost, need for support, gaining acceptance of new techniques – learning from past land degradation (fertile crescent, US mid-west) • **Judgement:** no single strategy sufficient – combination of ecological principles and new technology – justified conclusion **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **salinisation** → salt building up in the soil / soil becoming salty • **biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	
	Present agriculture relies on inputs that cannot last: chemical pesticides to which pests evolve resistance and which harm non-target species, rock phosphate that is non-renewable, nitrate fertiliser made with fossil-fuel energy, and irrigation that over-draws rivers and groundwater and salinises soil, while ploughing loses soil organic matter and livestock, rice and machinery release greenhouse gases. Each can be replaced by strategies based on ecological principles. Pests are controlled culturally by weeding, mulching, rotation, barrier crops and culling, by biological control and predator habitats, by polyculture and companion crops, and by integrated control that uses chemicals only as a last resort. Nutrients come from recycled organic matter, rotation with legumes and conserved soil biota, so artificial fertiliser, especially nitrate, can be reduced. Water is saved with low water-use crops, reservoirs and aquifer recharge and drip irrigation; soil carbon is kept by low tillage and less machinery, with renewable energy for what remains; methane falls with rice varieties that allow earlier drainage and with ground, carbohydrate-rich cattle feed. Habitats of crop wild relatives, seed banks, hedgerows, ditches, ponds and woodland and agri-environment schemes protect the biodiversity that supplies these services.	
	Food demand will rise with a population that may reach 12 billion, there is little new land, and imports can reduce food availability where it is grown, so yield cannot be ignored. Some strategies cost more or need farmers to accept new techniques, and past land degradation from the fertile crescent to the American mid-west shows what happens when they are not adopted. No single strategy suffices; a combination of ecological principles with new technology is the only route to feeding more people without exhausting the resources food depends on.	
	Weaker answer - Level 1 (2 marks)	
	To make agriculture sustainable farmers should use fewer chemicals. Organic farming does not use pesticides or artificial fertilisers. Crop rotation and manure keep the soil fertile. Farmers should not waste water. Hedges should be kept for wildlife. This would make farming more sustainable but it might produce less food.	
	What makes the difference	
	• Level 1 gives general advice; Level 3 identifies each unsustainable feature and matches it with specific strategies, showing understanding of why they work.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3's evaluation considers population growth, land limits, cost, adoption and historical evidence; Level 1's is a single clause.	
	• Note that Level 3 avoids defining organic farming as 'not using chemicals' - the textbook stresses it is the positive use of natural processes.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	
11.1	**Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Temperate broadleaf woodland:** seasonality, deep soils, high biodiversity, resources, climate and soil regulation – deforestation, fragmentation, management change – SSSIs, ancient woodland protection, coppicing, mixed planting • **Tropical rainforest:** stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils – fuelwood, timber, agriculture, mining, reservoirs, climate change – protected areas, debt-for-nature swaps, sustainable exploitation • **Tropical coral reefs:** cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity – fisheries, erosion protection, medicines, tourism – damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species – MPAs, no-take zones, control of activities • **Deep-water reefs:** cold, dark, slow growth – research, fisheries – trawling, oil / gas, acidification – SACs • **Oceanic islands:** isolation, endemism, no mammal predators	25

Do not write outside the box

Question	Marking guidance	Marks
	– exploitation, introduced species, habitat change, sea level – eradications, control of development and visitors • **Mangroves:** halophytes, anaerobic mud – erosion protection, fisheries nurseries, sediment trapping – clearance for aquaculture / urban / ports, pollution, climate change – protection, replanting • **Antarctica:** extreme cold, seasonality, marine nutrients, ice – water store, albedo, carbon, research – climate change, ozone, tourism, fishing, minerals – Antarctic Treaty, CCAMLR, tourism and waste rules • **Synthesis:** features -> importance -> threats -> strategy – evaluation of success – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	

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Question	Marking guidance	Marks
11.2	**Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Why monitor:** species present, population size, density, distribution, survival, age structure, breeding rates, movements – baseline and evaluation – Red List categorisation and quotas rely on it • **Standard techniques and their limitations:** quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR – sampling design (random / systematic, size, number, timing, standardisation) • **New technologies:** satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • **How they improve validity:** representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • **Examples:** penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked	25
	• **Limitations:** cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	

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